# Functional Gradients and Excitation–Inhibition Balance in the Early Developing Brain

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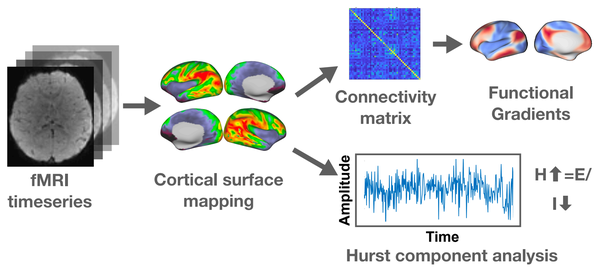
## Background & Challenges

- A clear characterization of large-scale functional gradients in the infant brain remains lacking, due in part to limited datasets and challenges in analyzing infant fMRI data.
- How excitation–inhibition balance contributes to the development of functional gradients during infancy is still not well understood.

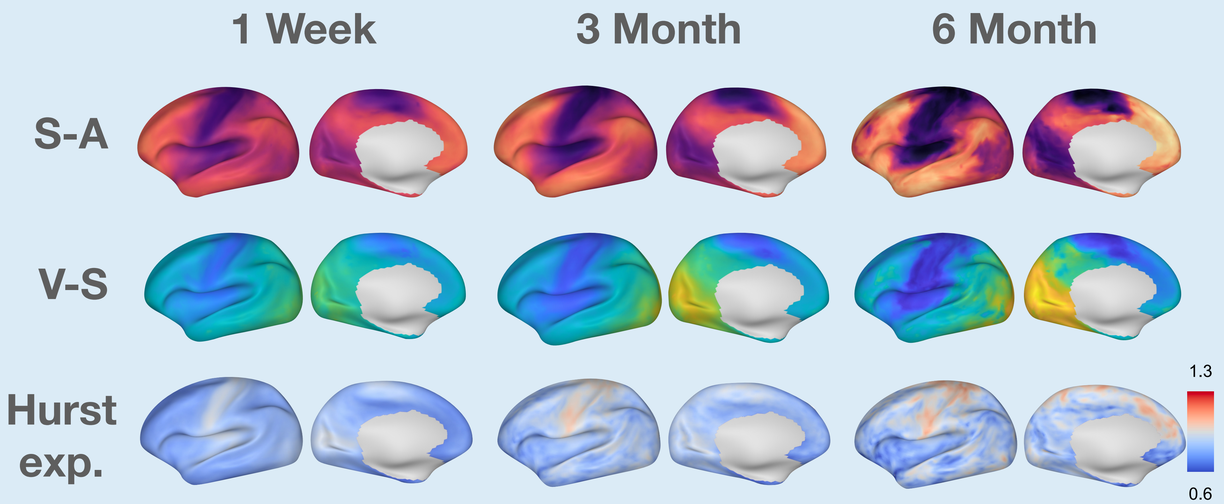
## Aims

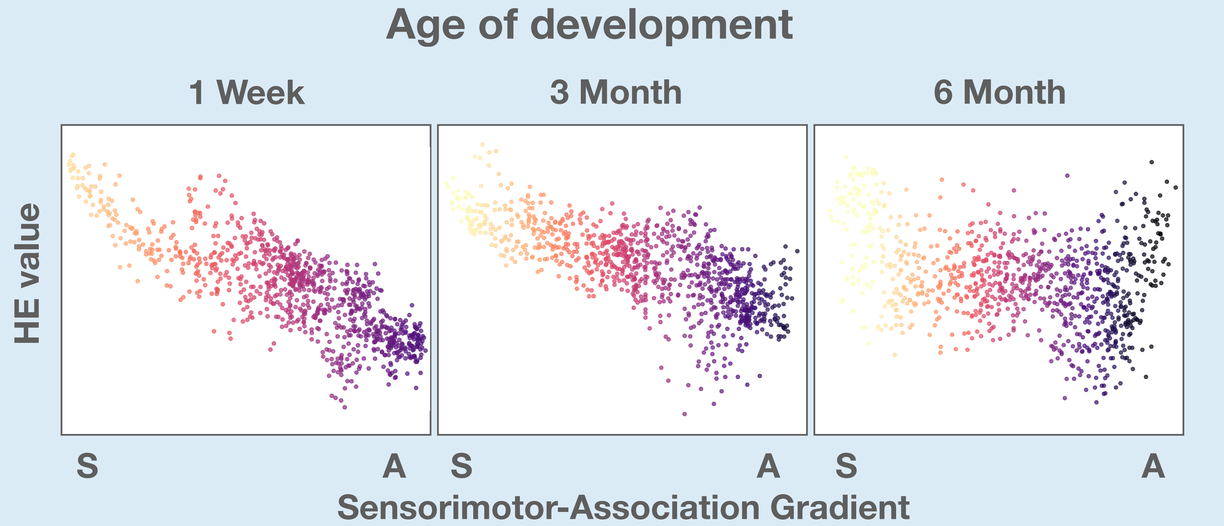
- Analyze vertex-level functional gradient of the infant brain, in relation to Hurst exponent, a marker of excitation-inhibition balance.
- Characterize cortical organization and its underlying dynamics in first 6 months of life.

## Methods



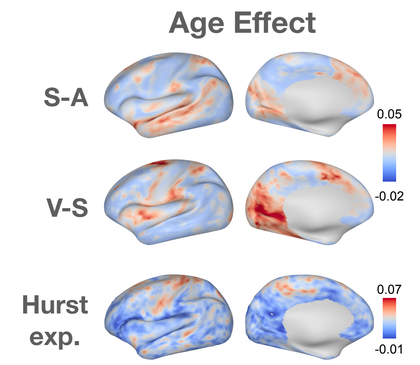

- Over 900 scans from 466 infants aged 0–6 months in Healthy Brain and Child Development (HBCD) dataset.
- Diffusion map embedding applied to top 10% strongest connections of each vertex, aligned by Procrustes alignment.
- Heightened E:I ratio links to reduced Hurst exponent of a BOLD time series.
- Generalized Additive Mixed Model (GAMM) for longitudinal modeling.

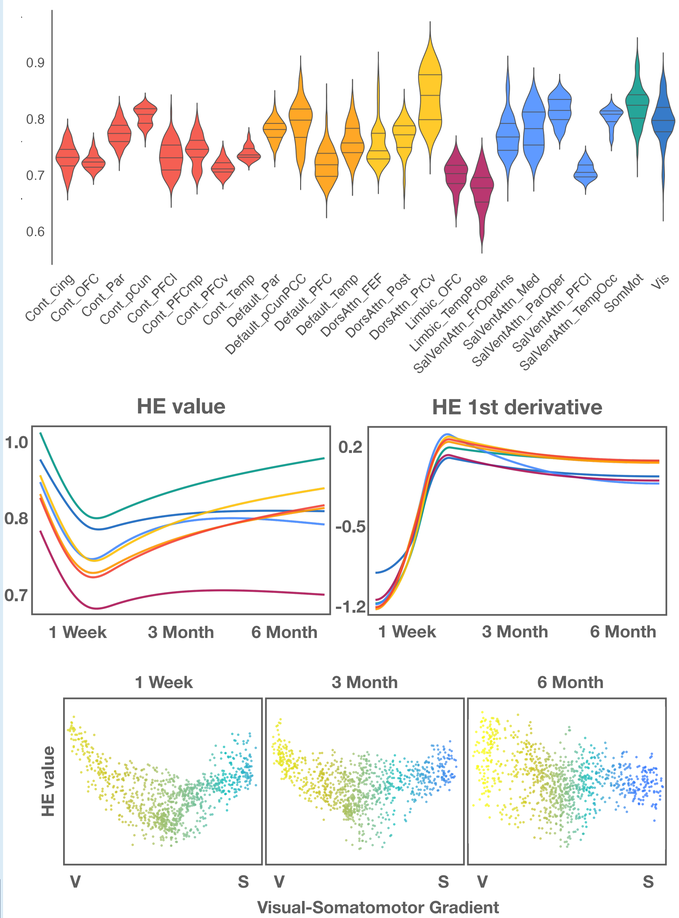



**Greater functional differentiation is associated with E/I imbalance in the first 3 months of life**

## Results




## References

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